Request for Proposals (RFP)

Open-H-Endoluminal Collaborative Endoluminal and Interventional Robotics Dataset Initiative

1 Introduction

Open-H-Endoluminal is a collaborative dataset generation and collection effort led by NVIDIA, Johns Hopkins University, Rice University, Bookout Center at Houston Methodist Hospital , and the Technical University of Munich, together with partner institutions, focused on endoscopic, endoluminal, and interventional robotics. Our goal is to align leading institutions to assemble real and synthetic training episodes that will advance endoluminal and interventional robotics. Data submitted to this effort will consist of high-quality video sequences synchronized with high-utility labels, such as robot state and action, and/or language across these procedures. The resulting datasets will be used to train and evaluate vision-language-action models, world models, and core perception capabilities such as depth estimation, 3D reconstruction, segmentation, and workflow understanding. All data from the initiative will be released CC BY 4.0 to enable the greater community to benefit from and build on permissive multi-embodiment endoluminal robotics datasets.

2 Steering Group

Role	Name	Affiliation
Industry Lead	Dr. Mahdi Azizian	NVIDIA
Academic Co-Lead	Prof. Mathias Unberath	Johns Hopkins University
Academic Co-Lead	Prof. Farshid Alambeigi	The Bookout Center at Houston Methodist Hospital / Rice University
Academic Co-Lead	Prof. Nassir Navab	Technical University of Munich

3 Scope of Work

Participating teams may submit for one or more of the following domains. Data is accepted broadly across the endoluminal scope.

3.1 Lower and Upper GI Endoscopy

Tasks include navigation, screening, detection, and intervention (for example biopsy, polypectomy, or wound sealing).

3.2 Bronchoscopy

Navigation to airway targets and transbronchial biopsy and ablation.

3.3 Endovascular and Catheter-Based Intervention

Navigation and interventional tasks across robotic endovascular and catheter-based procedures (for example thrombectomy, stenting, angioplasty, cardiac ablation, etc.).

3.4 Ureteroscopy and Transurethral

Navigation and diagnostic or interventional tasks in the urinary tract.

3.5 Wireless and Untethered Endoluminal Systems

Active locomotion, magnetic actuation, and navigation for capsule endoscopes and untethered robots in GI or other lumens not served by tethered platforms above

Out of scope: laparoscopy, arthroscopy, and rigid-instrument manipulation.

3.6 Data Requirements and Contribution Tiers

Each proposal must commit to a minimum data contribution and to providing time-aligned streams and required metadata.

Action and state signal. Submissions are graded into three signal tiers by the fidelity of their action and state signal. Label which you provide.

Signal Tier	Action / State Signal	Examples	Signal Weight
S1	Native robot kinematics	Joint and actuator state, motor commands, insertion depth, tip pose, teleoperation commands	×5
S2	Tracked pose	Electromagnetic (EM) tracking, fiber-optic shape sensing, EM shape sensing, magnetic tracker through the tool channel	×4
S3	Inferred pose	Pose inferred from camera (SLAM, SfM, point tracking) or from fluoroscopy	×2
S4	None (rich annotations)	No kinematics, instead high quality annotations are provided in the form of segmentation, depth or 3D, procedure phase, VQA, chain-of-thought, anomaly annotations, or de-identified reports	×2

Note: Datasets including only raw, unlabeled video are not accepted.

Time-aligned streams. Provide, synchronized: endoscopic video (RGB or RGB-D) and, where relevant, fluoroscopic or X-ray video; an action or state signal from the table above; and corresponding imaging where available, including CT and CT-fluoro pairs; text labels that describe the action being performed (frame-level labels > episode-level labels).

Format. Deliver data in the LeRobot data format. Where a robot is involved, supply calibration data and scope or robot CAD and kinematic-tree descriptions (USD, URDF, DH parameters, or equivalent).

Required metadata (every submission). Task intent (navigation, screening, detection, or intervention) and the target; device or platform; collection setting (in-vivo, ex-vivo, phantom, or simulation); modalities present with their synchronization and sample rates; and licence and de-identification status.

Fluoroscopy dataset minimum contribution. For fluoroscopy-based datasets, please directly inquire with your estimated dataset duration, number of unique sequences and/or clips, number of distinct anatomical/phantom vessel morphologies (both overall and distinct paths within the same model), and other details as deemed informative, because fluoroscopy dataset proposals will be considered on a case-by-case basis due to the more unique context and constraints. The "Collection Setting" and "Signal Tiers" below still apply and should be clarified in any proposal.

RGB endoscopy datasets minimum contribution. The below minimums are specified in hours of synchronized data per collection setting and pertain only to RGB endoscopy datasets. The S1 column is the base requirement; because S2 and S3 signals carry proportionally lower weight, their minimums scale up so that each tier represents an equivalent contribution. Preferred order of collection setting: clinical, in-vivo, ex-vivo, phantom or bench-top, simulation.

Collection Setting	Definition	S1 (Native)	S2 (Tracked)	S3 (Inferred)	S4 (Annotations)
Clinical	Collected during approved human procedures with appropriate IRB or ethics oversight.	2 h	3 h	5 h	5 h
In-Vivo	Acquired on living tissue (for example a porcine model) under laboratory or veterinary supervision.	2 h	3 h	5 h	5 h
Ex-Vivo	Performed on isolated tissue samples (for example ex-vivo bowel or airway).	4 h	6 h	10 h	10 h
Phantom / Bench-Top	Conducted on physical phantoms, synthetic tissues, or bench-top fixtures.	8 h	12 h	20 h	20 h
Simulation (Digital)	Digitally simulated using physics-based or differentiable environments.	20 h	30 h	50 h	50 h

Data Collection and Manipulation Diversity Requirements. To promote robust learning and minimize action-distribution bias, RGB endoscopy datasets should include a balanced distribution of manipulation strategies. As a target, approximately half of the collected trajectories should primarily involve translational motions, such as insertion and retraction, while the remaining trajectories should require active articulation, steering, bending, or coordinated combinations of articulation and translation to navigate the workspace or complete the task. Data should span diverse workspace geometries and configurations to provide broad state-action coverage and prevent models from overemphasizing translational motion at the expense of effective steering and manipulation.

For colonoscopy, colon phantoms or simulated anatomies should be configured across a range of bent, looped, and tortuous geometries rather than remaining predominantly straight, so that navigation requires meaningful steering and coordinated insertion. The same principle applies to other flexible endoscopic and robotic systems: airway, upper-GI, urinary-tract, vascular, and other relevant anatomical or phantom geometries should be varied to elicit representative use of the platform's translational and articulation degrees of freedom. Proposals should describe the planned distribution of translation-dominant and articulation-requiring trajectories, the geometry or anatomy variations used to achieve it, and how manipulation strategy and configuration will be captured in the dataset metadata.

Additional metadata. We encourage proposals to also provide: time-aligned narration or description of the sub-task (audio or text); correlated anonymized patient information where available; camera intrinsics; and labels for demonstration quality (expert, intermediate, novice) and task success (failure, recovery, success).

3.7 Focused Tracks and Alternative Formats

To capture the benefits of focused, high-quality datasets while keeping the initiative flexible, we invite groups to propose focused tracks around specific areas, for example lower-GI intubation on a shared reference phantom, polypectomy or biopsy through the tool channel, or endovascular navigation in simulation. We also welcome standardized pre-calibration and shared-phantom protocols and alternative data formats.

3.8 Reference Platforms

For groups interested in participating but lacking existing hardware, we encourage adopting the OpenRC framework to enable low-cost robotics research within lower GI. If interested in adopting this framework, please contact: Siddhartha Kapuria - skapuria@utexas.edu.

4 Timeline

Milestone	Date
RFP released	July 2026
Proposal submission deadline	September 2026
Data collection window	September 2026 to January 2027
Data sample deadline	December 1, 2026
Data cleanup and standardization	January 2027
Model training and validation	February 2027
Public release of dataset and models	March 2027

5 Eligibility and Partnership Models

- Academic institutions, start-ups, and industrial healthcare firms are welcome.
- International participation is encouraged.
- Consortia of multiple organisations may submit a joint proposal with a single lead.

Collaboration benefits. All accepted collaborators will be named co-authors on the Open-H-Endoluminal dataset publication and on the follow-up paper describing the resulting models. Teams will receive early-access evaluation checkpoints of the models and the dataset itself for their own downstream experiments. Participating teams will also receive recognition during the dataset and model releases at NVIDIA GTC 2027.

6 Proposal Requirements

Submissions (single PDF, ≤ 5 pages excluding appendices) must include:

1. Executive summary (≤ 500 words).
2. Task and domain selection: target domain or domains from Section 3, tasks along the hierarchy, device or platform, which signal tier (S1, S2, or S3) you provide, sensors, and committed hours of data per collection setting.
3. Technical approach: data-collection methodology, synchronization, quality assurance, and privacy safeguards (HIPAA and GDPR).
4. Data diversity strategy: describe how the data accounts for diversity, such as the skill levels of operators, variations in devices used, lighting conditions, patient variability, or phantom model configurations.
5. Team qualifications: key personnel biographies and prior related work.
6. Project plan: work-breakdown structure and Gantt chart aligned with the Section 4 timeline.
7. Data rights and IP statement: confirmation of intent to release data under the Creative Commons CC BY 4.0 licence (or compatible) and to comply with steering-group IP policy.

Submit proposals to the [Open-H-Endoluminal Google Form].

7 Data Governance, Ethics, and Compliance

- **Patient privacy.** All human-subject data must be de-identified to HIPAA Safe Harbor standards or equivalent (GDPR for European contributors), including scrubbing DICOM and imaging metadata on CT, fluoroscopy, and video exports.
- **Regulatory.** Proposers are responsible for obtaining IRB or ethics approval where required.
- **Licensing.** Final datasets will be released under CC BY 4.0. Synthetic assets must be free of third-party IP encumbrances. Composers are responsible for obtaining IRB or ethics approval where required.

- **Release.** Contributors accept that their datasets will be uploaded publicly to HuggingFace.

8 Contact Information

- **Technical questions:** Nigel Nelson, nigeln@nvidia.com (NVIDIA); Siddhartha Kapuria, skapuria@utexas.edu (UT Austin)
- **Administrative questions:** Sean Huver, shuver@nvidia.com